RIVER VALLEY HIGH SCHOOL
YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME					
CENTRE NUMBER		INDEX NUMBER		CLASS	

H1 BIOLOGY **8875/02**

Paper 2 **20 September 2013**

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.
Circle the question number in the table on the right.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	**/ 9**
2	**/ 9**
3	**/ 8**
4	**/ 5**
5	**/ 9**
6 or 7*	**/ 20**
Total	**/ 60**

This Question Paper consists of **13** printed pages.

Section A

Answer **all** the questions in this section.

1. **Fig. 1.1** is a diagram of an electron micrograph of part of a plant cell.

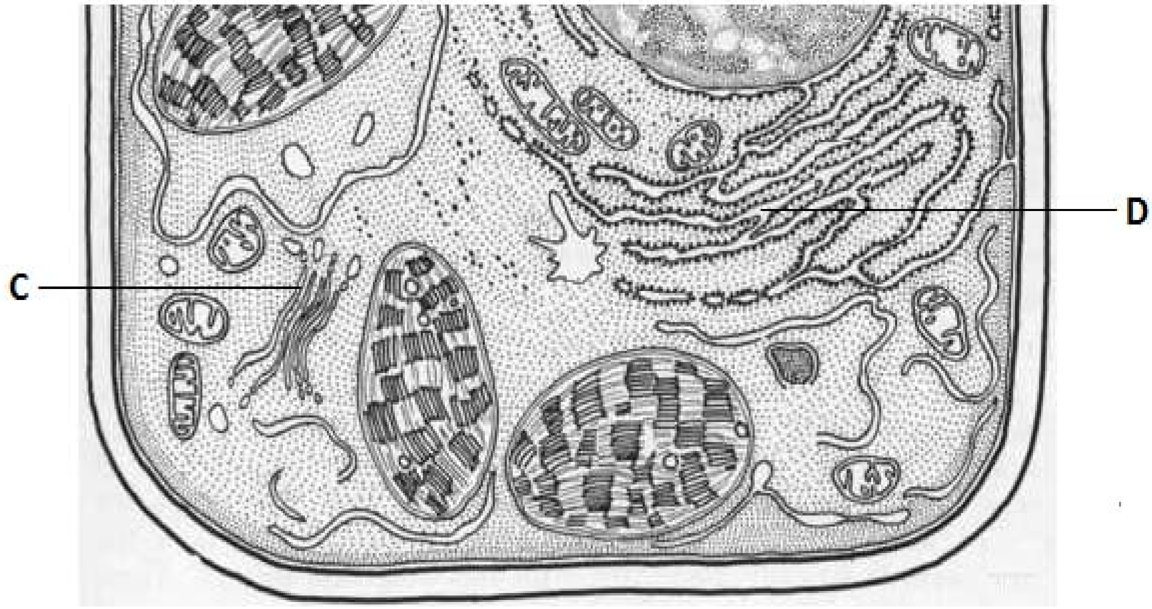

Fig. 1.1

(a) Using lines, indicated precisely in **Fig. 1.1**, a possible location where each of the following molecules can be found:

A: chlorophyll

B: cellulose [2]

(b) **(i)** Identify the organelle labelled **C** and state a visible feature that enabled your identification. [2]

1. **(b)** **(ii)** Explain how the functions of the two organelles, **C** and **D**, are linked. [2]

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Fig. 1.2 shows a magnified imaged of organelle **D**.

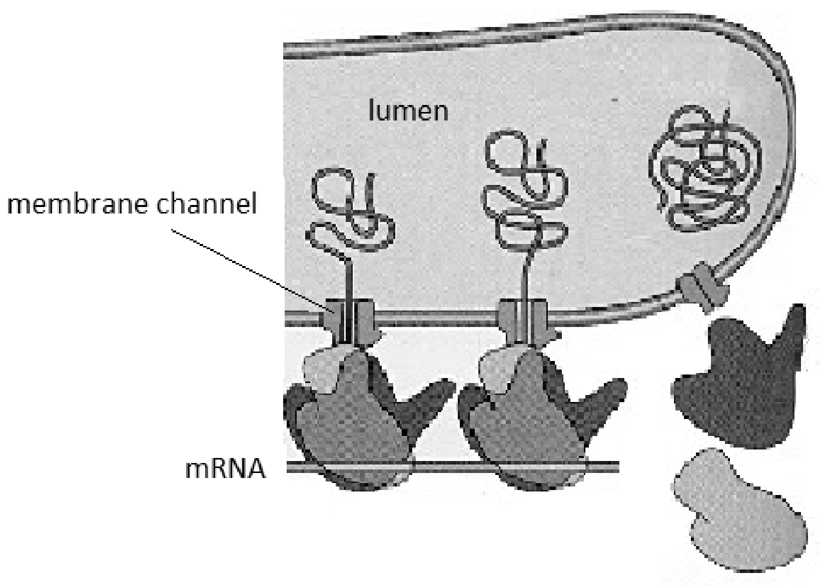

Fig. 1.2

(c) Describe the function of the protein that forms the membrane channel. [3]

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[Total: 9]

2. **PART I**

Fig. 2.1 is an electron micrograph of chromatin maintained at the 30 nm higher-order fiber.

Fig. 2.1

(a) Most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm.

(i) Describe how DNA is packed into the 30 nm chromatin fibre. [2]

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(ii) Explain why the genome is packaged this way. [2]

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2. **PART II**

Fig. 2.2 and **Fig. 2.3** are diagrams showing DNA replication and transcription, respectively.

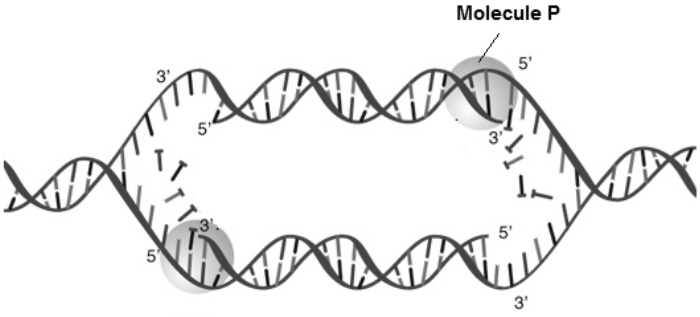

Fig. 2.2

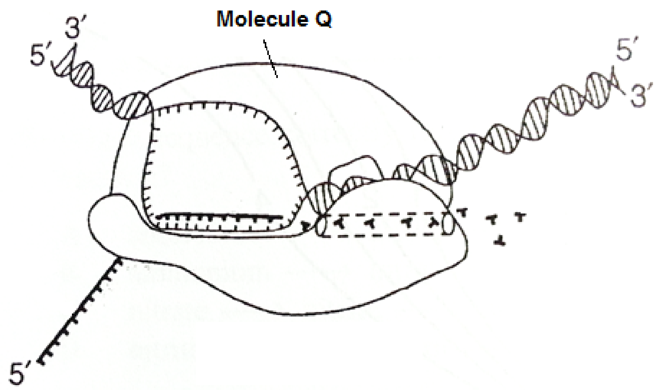

Fig. 2.3

(b) Explain why RNA primer is required for the synthesis of new strand in DNA replication. [1]

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(c) Describe **two** ways in which the functions of molecules **P** and **Q** are

(i) similar to each other; [2]

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2. **(c)** **(ii)** different from each other. [2]

[Total: 9]

3. **<u>PART I</u>**

Fig. 3.1 is a diagram of an electron micrograph of part of a chloroplast showing thylakoid membranes.

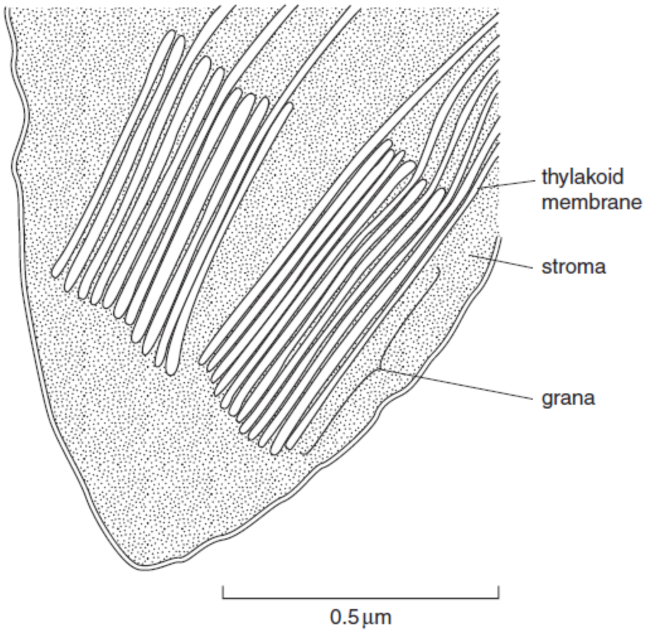

Fig. 3.1

(a) Describe the role of thylakoid membrane in photosynthesis. [4]

3.

PART II

Arabidopsis thaliana is a small flowering plant native to Asia. A mutation in the gene coding for NADP oxidase results in plants with short root hairs. NADP oxidase is an enzyme that converts NADPH to $NADP^+$. In an experiment, the length of root hairs in wild type and mutant *A. thaliana* were measured at intervals, together with the activity of NADP oxidase in the tips of the root hairs. The results are shown in **Fig. 3.2**.

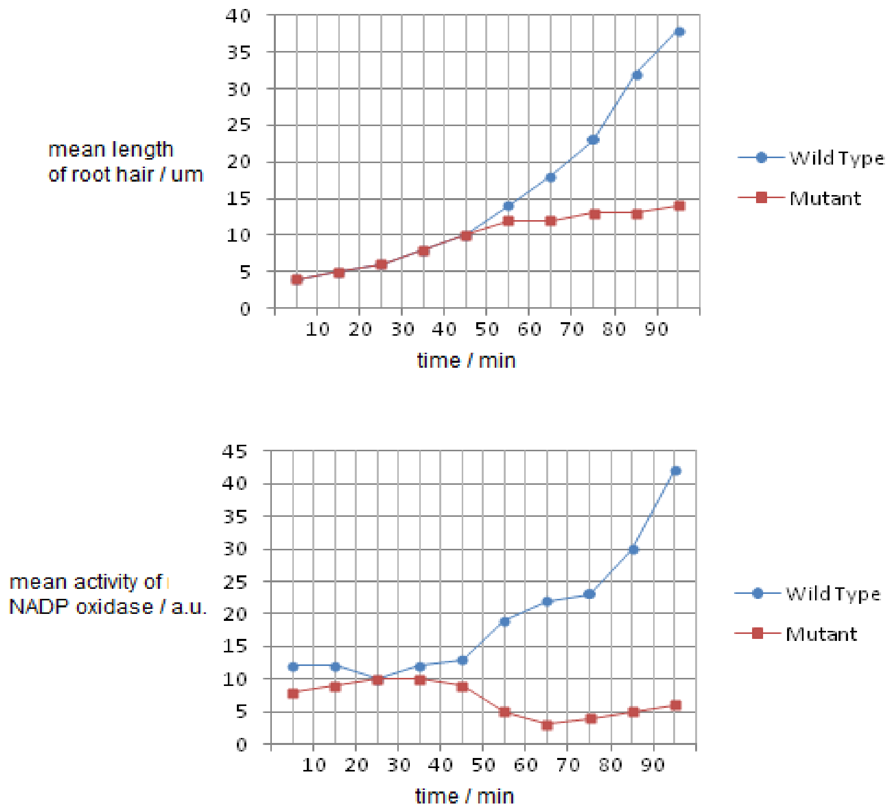

Fig. 3.2

(b) With reference to **Fig. 3.2**, and your knowledge of photosynthesis, describe and explain the difference in the growth of root hairs in the two types of *A. thaliana*. [4]

[Total: 8]

4. **Fig. 4.1** shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates.

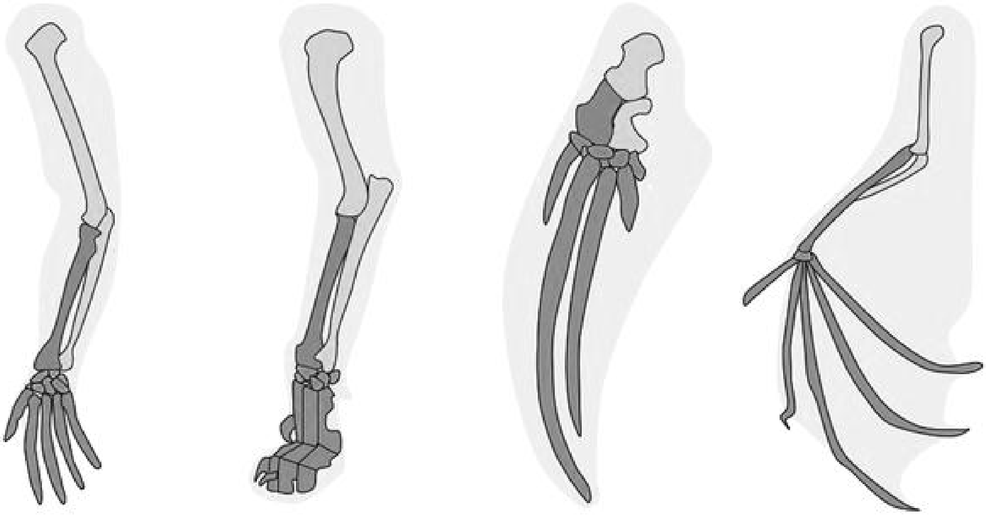

Fig. 4.1

(a) Explain how the relationship between the structures in **Fig. 4.1** provides evidence to support the theory of evolution. [2]

(b) Suggest how natural selection could have brought about the evolution of the skeleton of a bat's wing. [3]

[Total: 5]

5. **PART I**

DNA fingerprinting is a method of identification based on an individual's DNA. In traditional DNA fingerprinting, DNA is first isolated from cells, and then amplified by Polymerase Chain Reaction (PCR). **Fig. 5.1** illustrates the process of PCR.

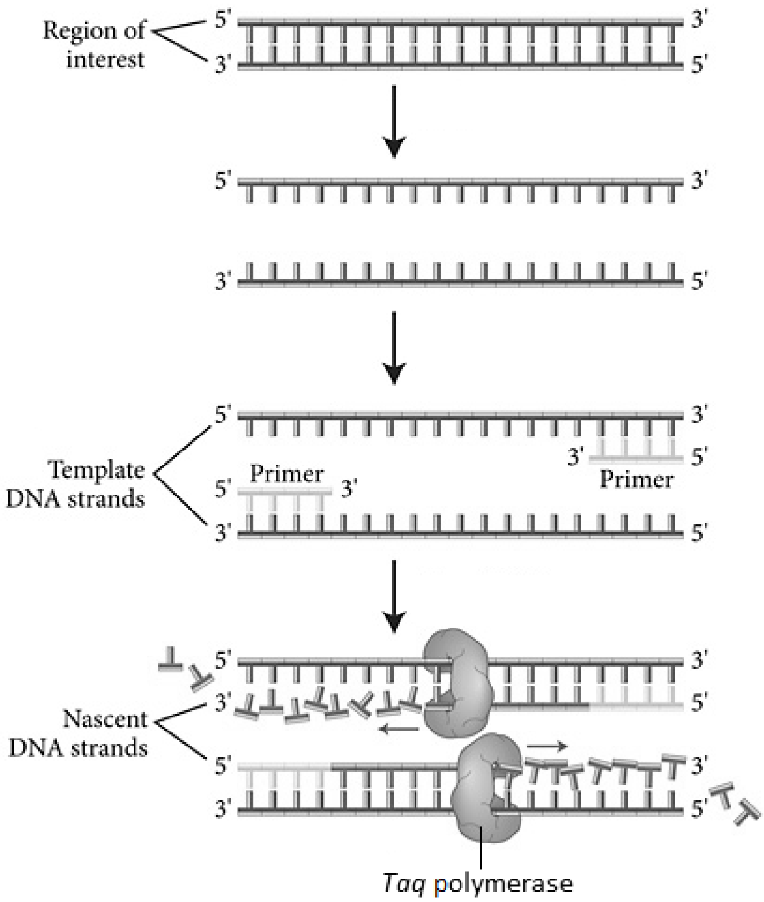

Fig. 5.1

(a) Explain how the specific nucleotide sequence of the primer is important for its role in PCR. [2]

5. **(b)** Distinguish between PCR and DNA replication in human skin cells. [2]

PART II

Rice is the staple diet in many parts of the world. Rice plants produce the precursor of vitamin A, β-carotene, in their green tissues, but not in the edible endosperms of their seeds. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β-carotene, was developed in Switzerland by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium. The rice produces seeds containing β-carotene.

Fig. 5.2 shows the DNA sequence used.

Key
pro - start site for polymerase enzymes
ter - end signal for polymerase enzymes
psy - gene from daffodil (a flowering plant)
crt 1 - gene from bacterium
Hyg resist - antibiotic resistance gene from a bacterium

Fig. 5.2

The following steps were taken to produce golden rice:

Step 1: a length of DNA (shown in **Fig. 5.2**) was made including a rice endosperm-specific promoter, *psy* gene, *crt* 1 gene and *Hyg resist* gene,

Step 2: copies of this length of DNA were inserted into plasmids from the bacterium *Agrobacterium tumefaciens*,

Step 3: *A. tumefaciens* containing the plasmids were mixed with rice embryos and cultured in a medium containing the antibiotic *Hyg*,

Step 4: the embryos growing successfully on *Hyg* were grown into plantlets and then plants.

5. **(c)** Explain how a length of DNA can be inserted into a plasmid in **Step 2**. [3]

(d) Explain the role of the *Hyg* resistance gene in this procedure. [1]

Field trials of genetically engineered crops, such as golden rice, hope to identify any risks to the environment or to human health of growing and eating the crops.

(e) Suggest one possible risk to the environment of growing a genetically engineered crop. [1]

[Total: 9]

Section B

Answer EITHER 6 OR 7.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

6. **(a)** Describe the structure of the cell membrane and relate it to its properties. [7]

(b) Explain the significance of mitosis in growth, repair and asexual reproduction. [4]

(c) Using named examples, explain the functions of stem cells in a living organism. [9]

[Total: 20]

7. **(a)** Distinguish between competitive and non-competitive inhibition of enzymes. [7]

(b) Describe how the Calvin cycle differs from the Krebs cycle. [8]

(c) The genetic manipulation of plants and animals, and their use in agriculture, is one the most controversial scientific development of recent times. What are some moral issues associated with the rise of such technology? [5]

[Total: 20]